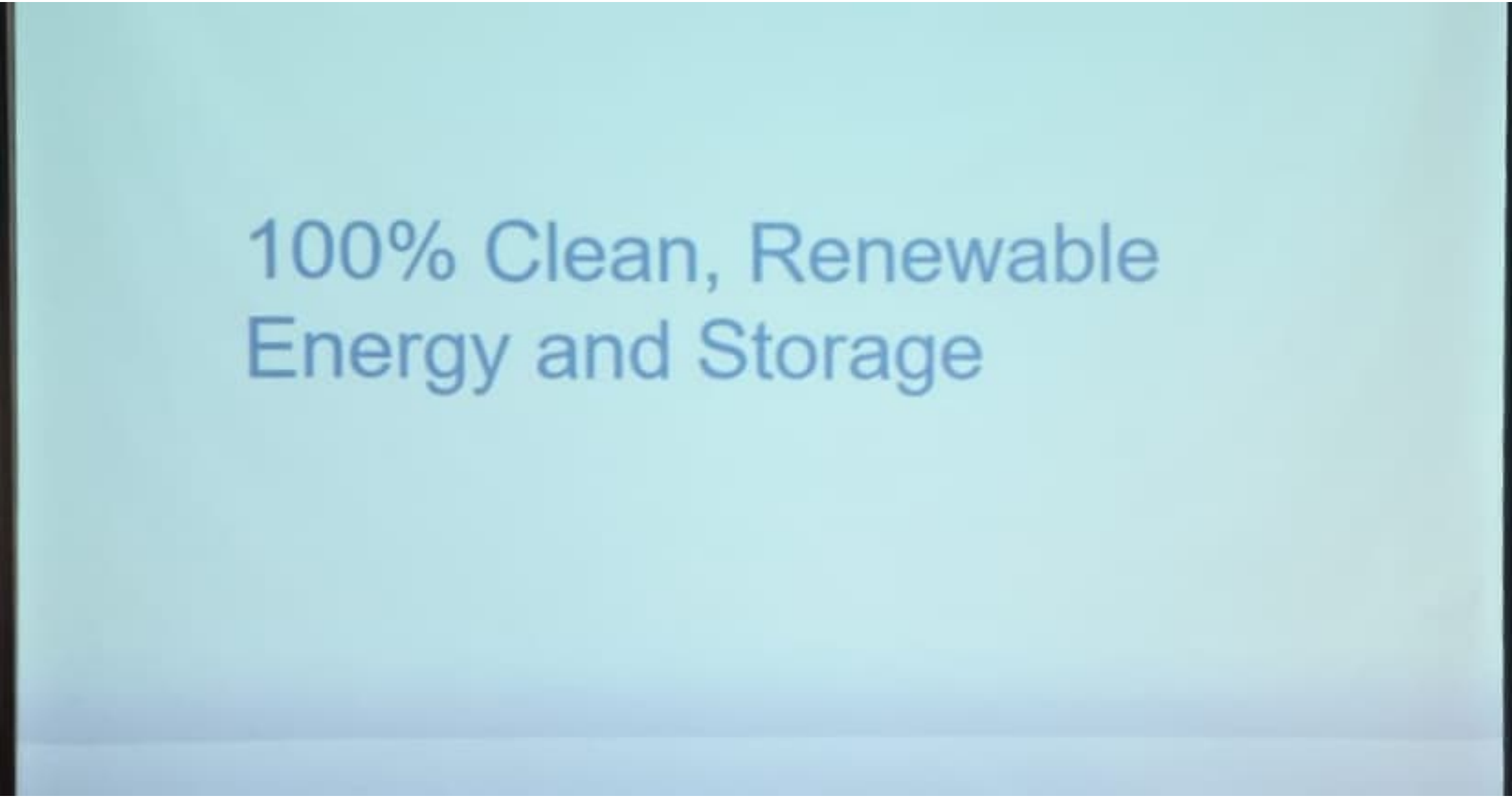

100% Clean, Renewable
Energy and Storage



Wind-Water-Solar and Storage Solution

Wind, Water, Solar (WWS) Solution

Electrify or Provide Direct Heat For All Sectors and Provide the Electricity and Heat with 100% WWS

ELECTRICITY	TRANSPORTATION	HEATING/COOLING	INDUSTRY
Wind	Battery-electric	Electric heat pumps	Electric arc furnaces
Solar PV/CSP	H_2 fuel cell	Solar heat	Induction furnaces
Geothermal		Geothermal heat	Resistance furnaces
Hydro		District heat/cold	Dielectric heaters
Tidal/Wave			Electron beam heaters

Onshore and Floating Offshore Wind





WWS Generation

WWS electricity generation
- Onshore/offshore wind
- Rooftop/utility photovoltaics
- Concentrated solar power
- Geothermal
- Hydro
- Tidal & wave

WWS heat generation
- Solar thermal/CSP steam
- Geothermal heat

WWS Grid

Transmission/distribution (T&D)
- AC/HVAC/HVDC lines
- Distribution lines
- Grid management
  - Software
  - Demand response

WWS Storage

Electricity storage
- Batteries
- CSP storage
- Pumped hydropower storage
- Hydropower reservoirs
- Flywheels
- Compressed air
- Gravitational storage

District heat storage
- Water tanks
- Boreholes/water pits/aquifers

District cold storage
- Water tanks/ice
- Aquifers

Building heat storage
- Water tank storage
- Thermal mass

Hydrogen storage
- Hydrogen storage tanks

WWS Equipment

Building and district air+water heating
- Electric heat pumps

Building and district cooling
- Electric heat pumps

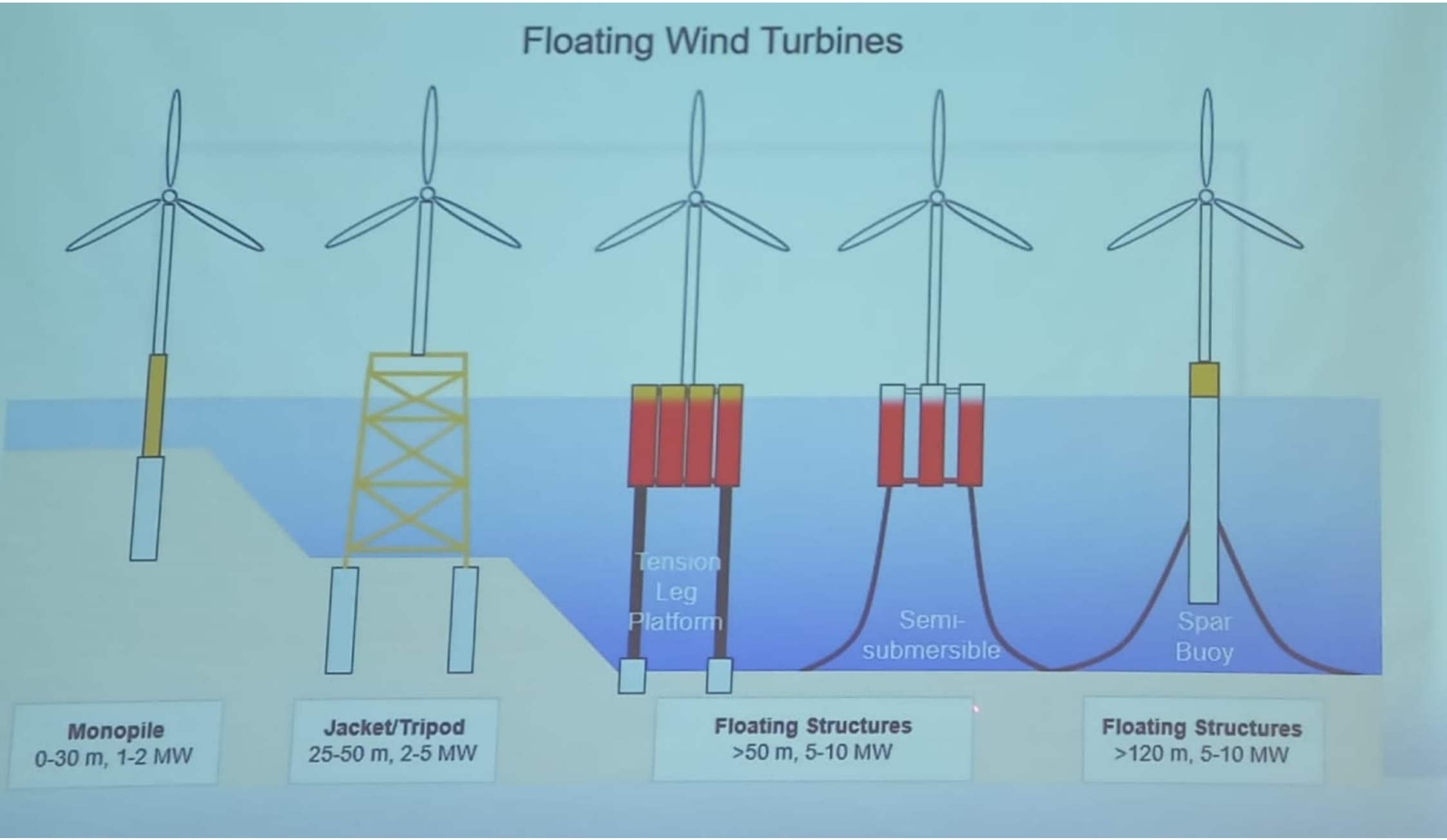
Industrial heat
- Arc/induction/resistance furnaces
- Dielectric heaters
- Electron beam heaters
- Heat pumps/CSP steam

Hydrogen generation/compression
- Electrolyzers/compressors

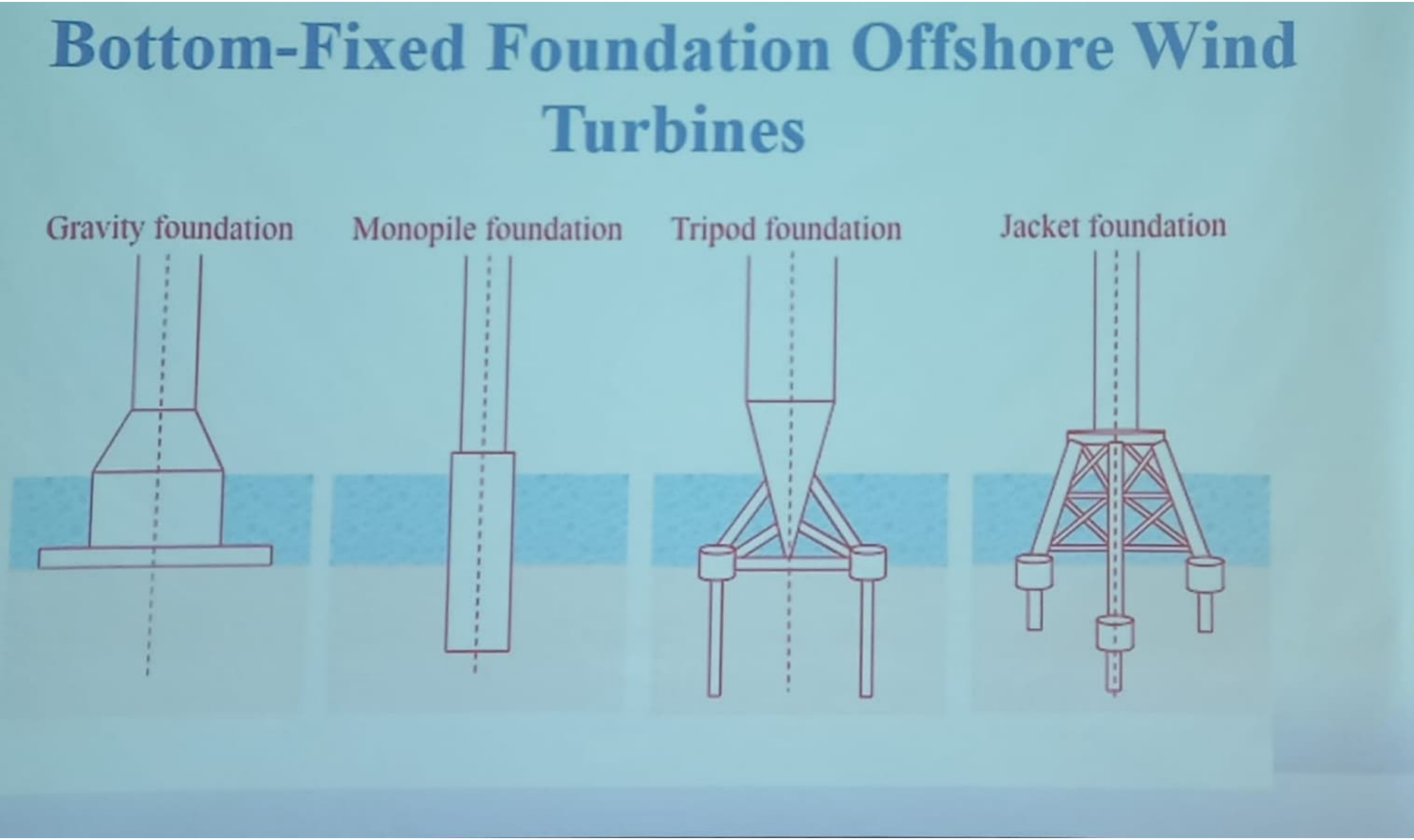
Transportation vehicles
- Battery-electric
- Hydrogen fuel cell

Some appliances/machines
- Induction cooktops
- Electric leafblowers/lawnmowers
- Heat pump dryers

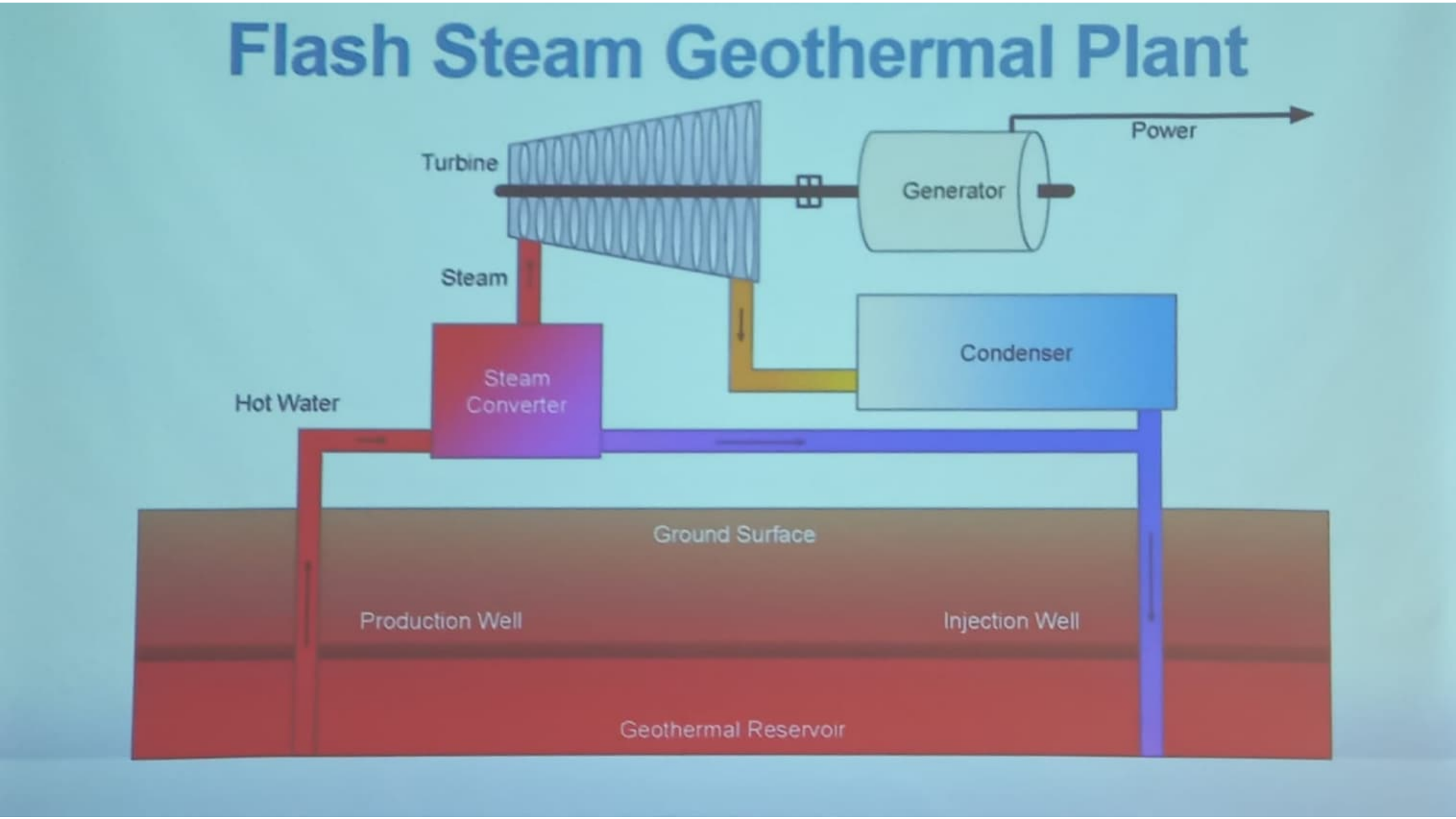
Efficiency/reduce energy use
- Insulate/weatherize buildings
- LED lights/efficient appliances
- Telecommute/improve public transit


Floating Wind Turbines
Tension Leg Platform
Semi-submersible
Spar Buoy
Monopile
0-30 m, 1-2 MW
Jacket/Tripod
25-50 m, 2-5 MW
Floating Structures
>50 m, 5-10 MW
Floating Structures
>120 m, 5-10 MW

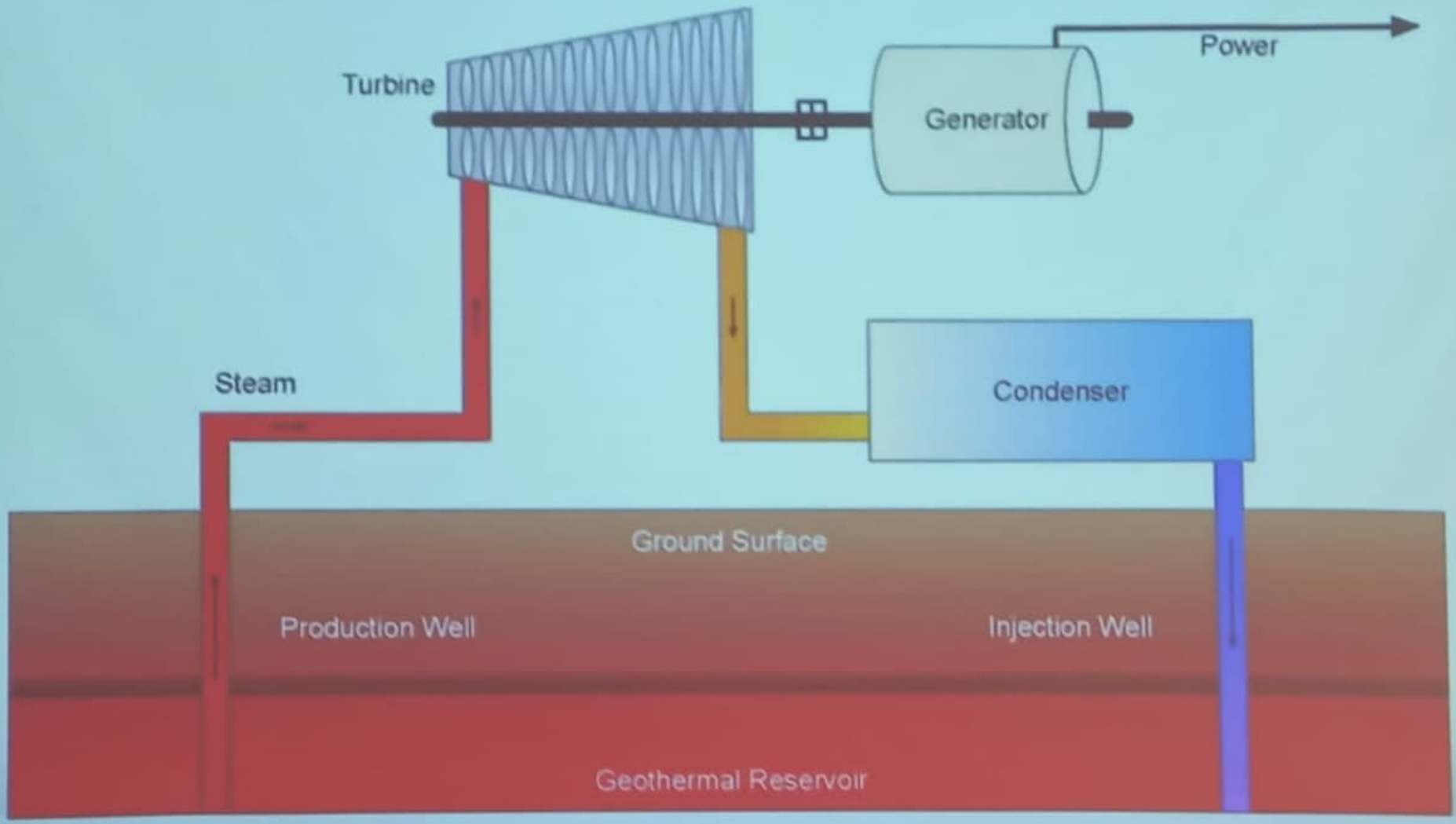
Bottom-Fixed Foundation Offshore Wind Turbines



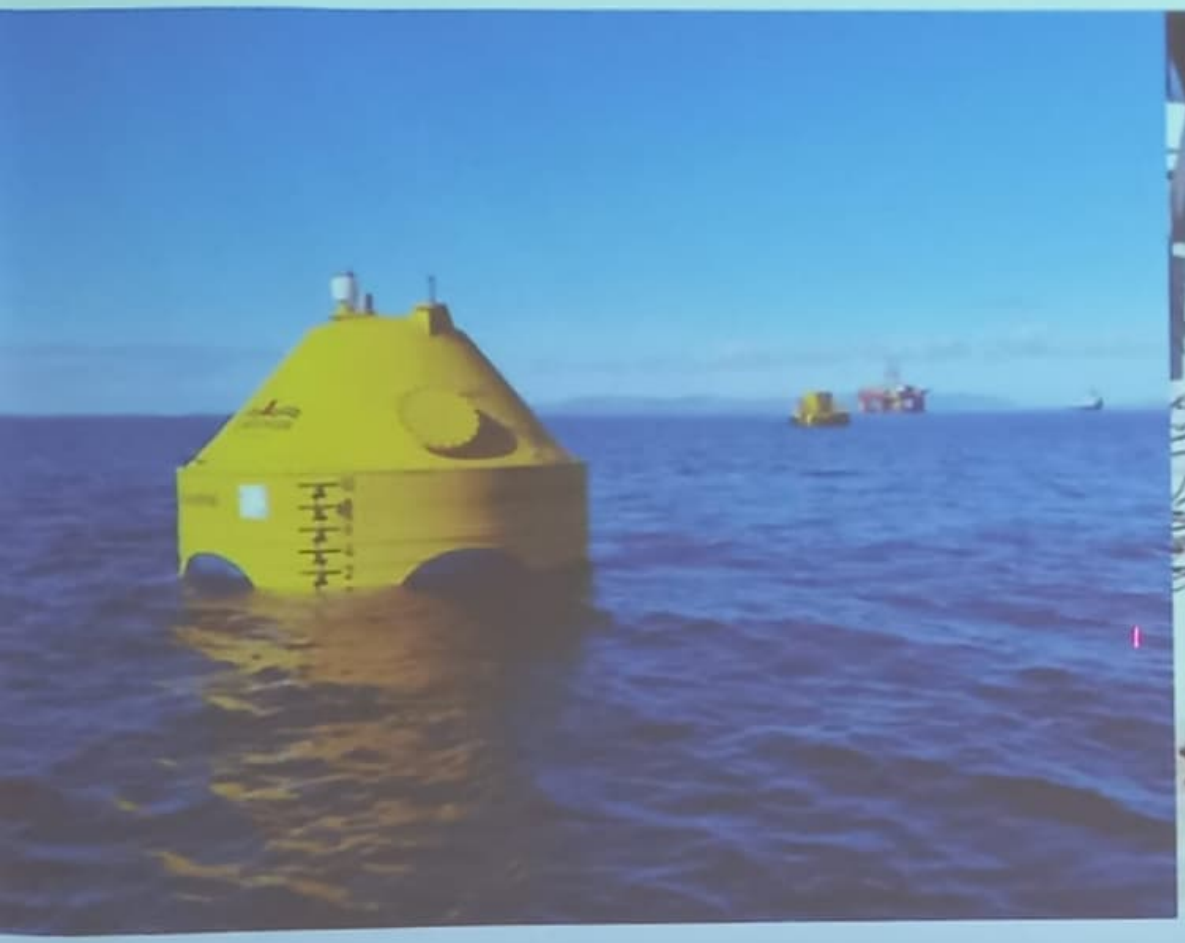
Flash Steam Geothermal Plant



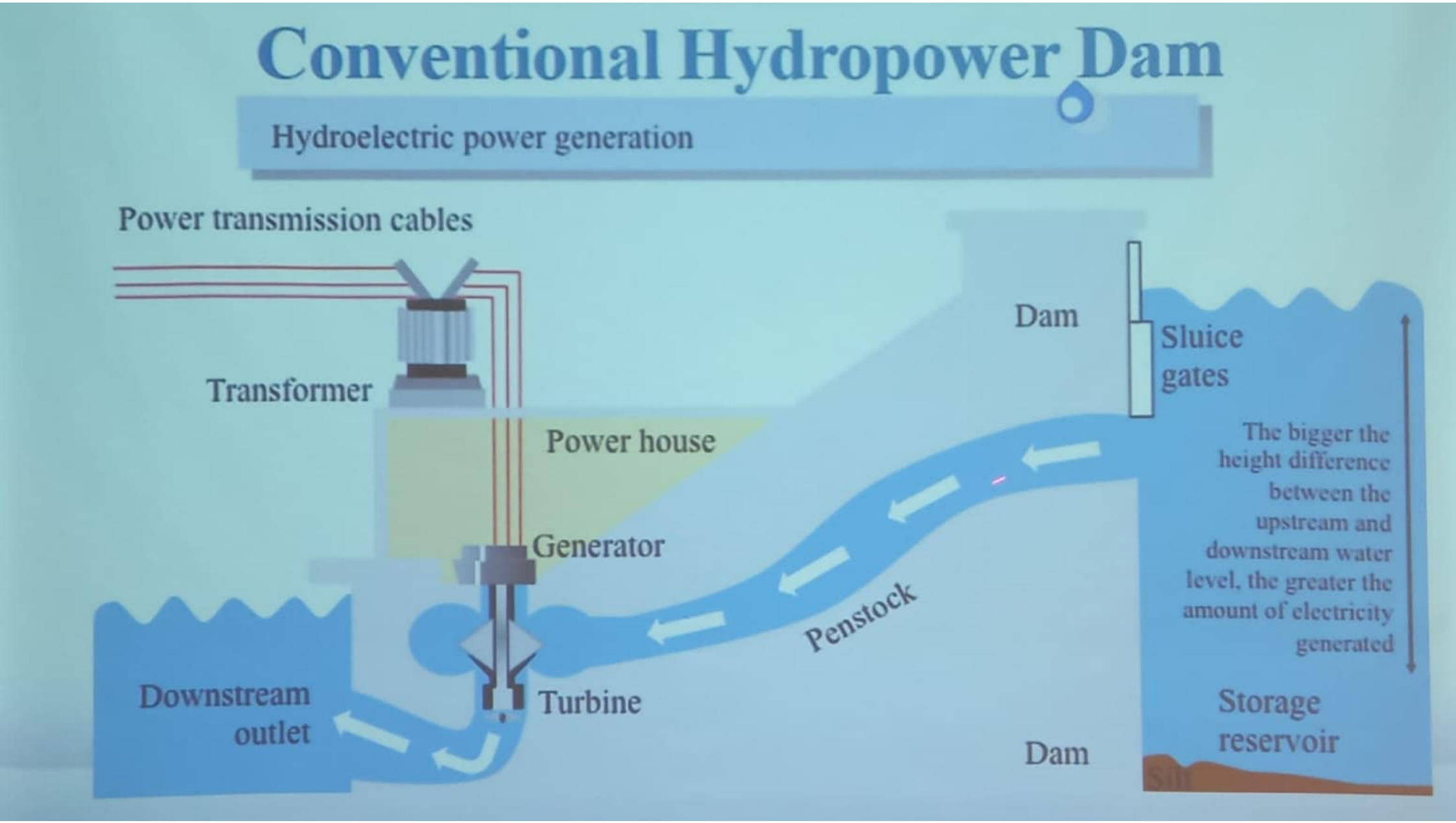
Dry Steam Geothermal Plant



Free-Floating and Land-Fixed Wave Devices





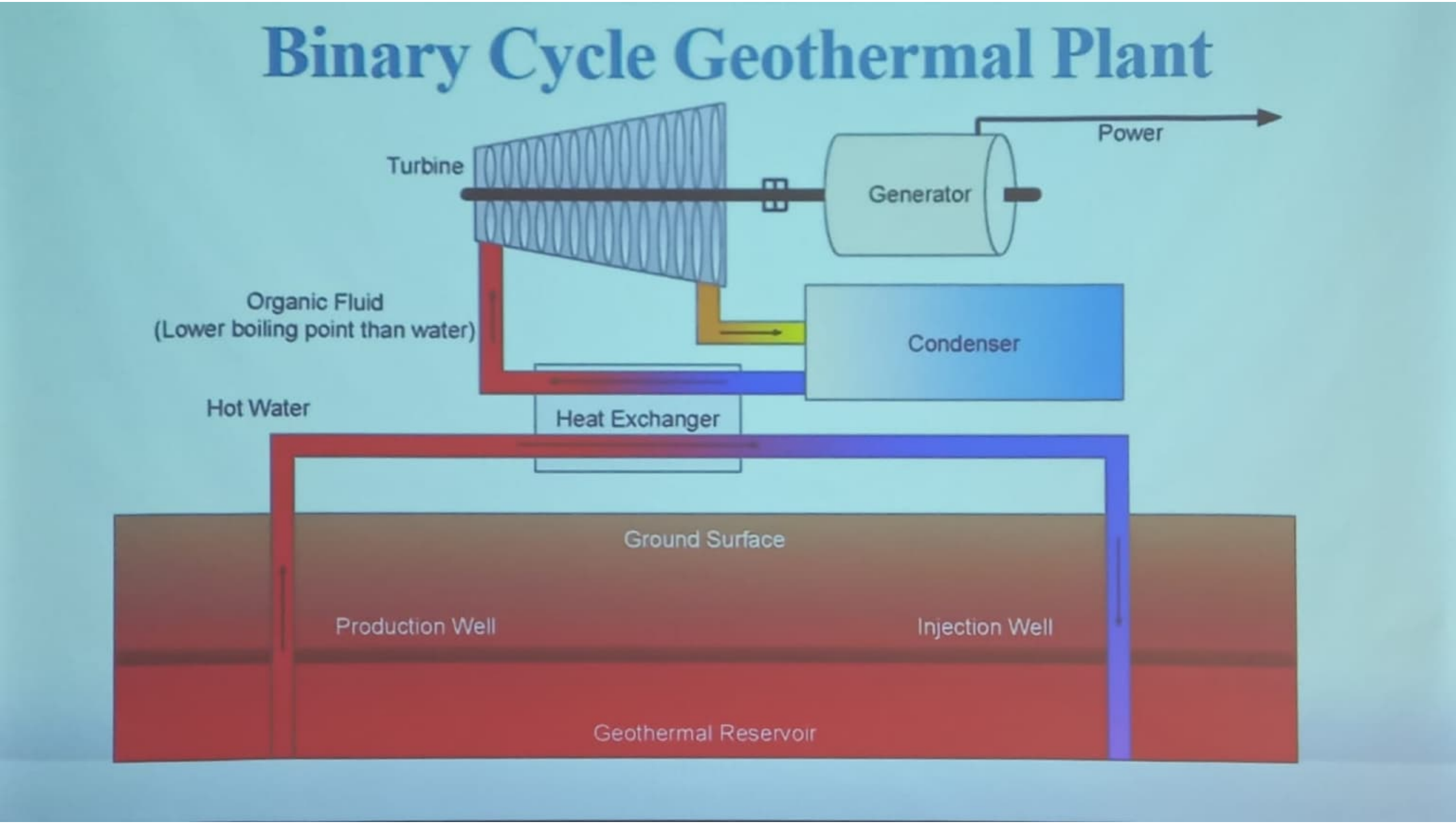

Conventional Hydropower Dam
Hydroelectric power generation
Power transmission cables
Transformer
Power house
Generator
Turbine
Penstock
Dam
Sluice gates
The bigger the height difference between the upstream and downstream water level, the greater the amount of electricity generated
Storage reservoir
Dam
Downstream outlet

Tidal Turbines

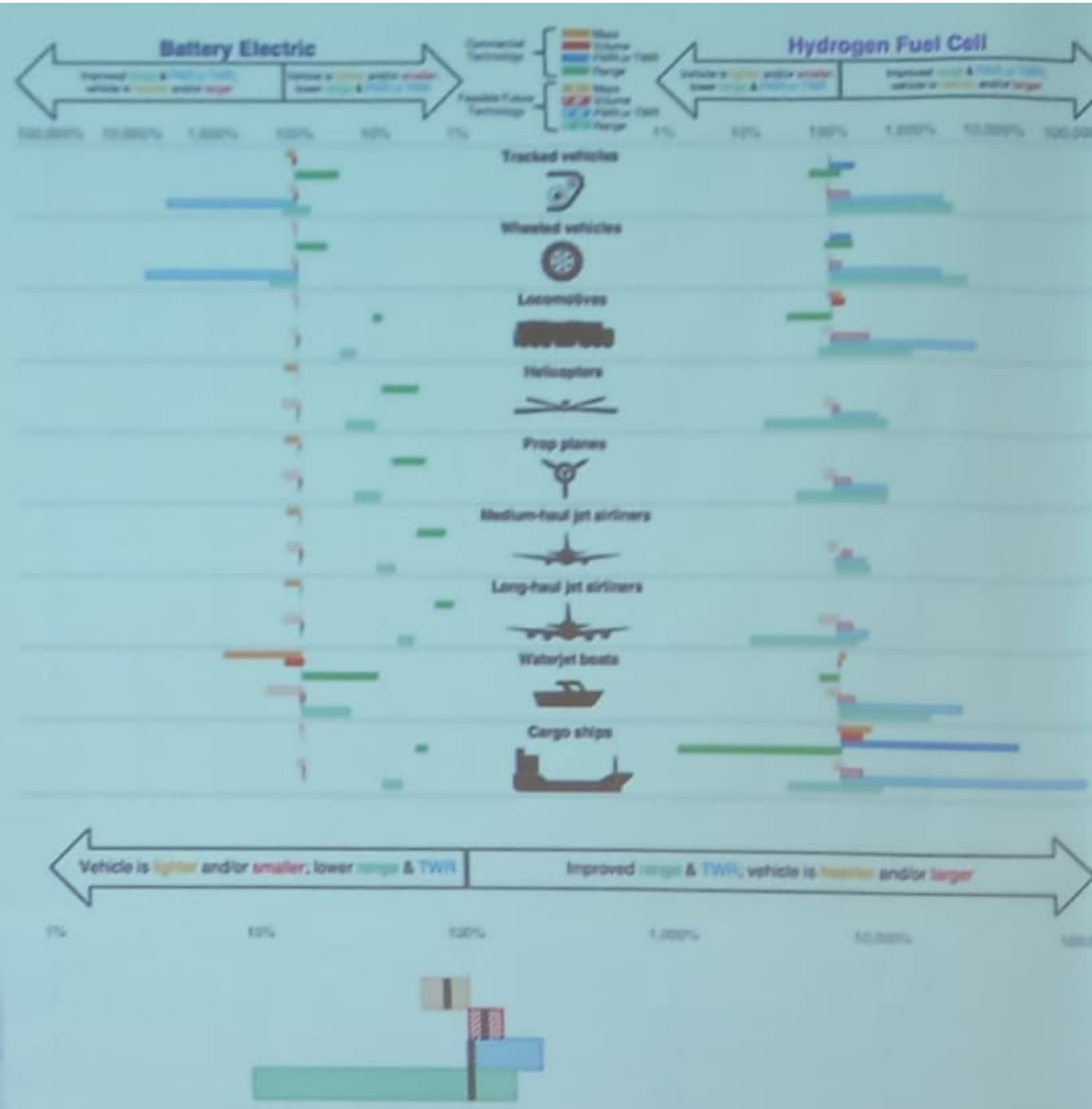




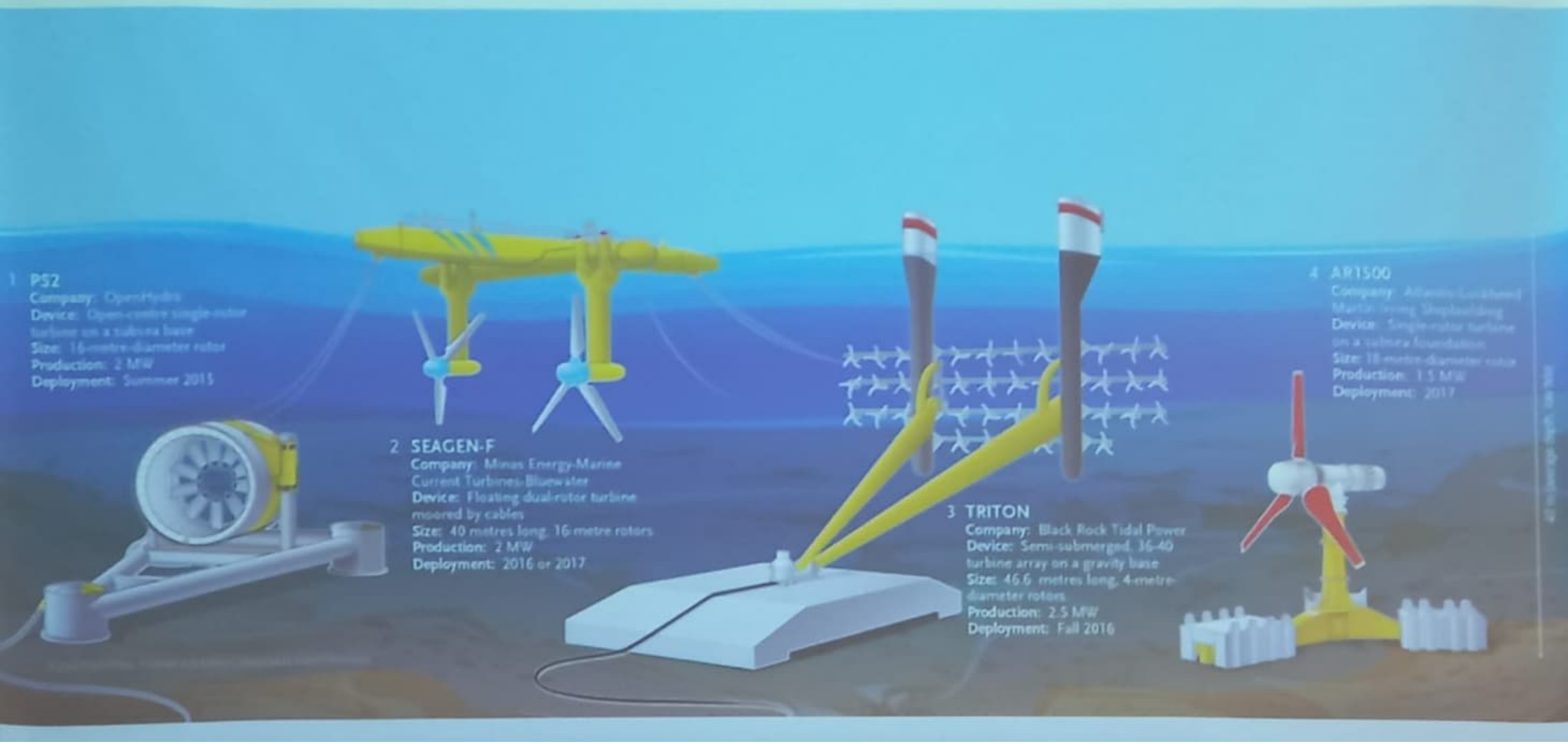
Binary Cycle Geothermal Plant



Percent changes in mass, volume, and range upon converting from an ICE military or civilian vehicle to a BE or HFC vehicle assuming same or better power (or thrust)-to-weight ratio



In-Stream Tidal Turbines



1 PS2
Company: OpenHydro
Device: Open-centre single-rotor turbine on a subsea base
Size: 16-metre-diameter rotor
Production: 2 MW
Deployment: Summer 2015

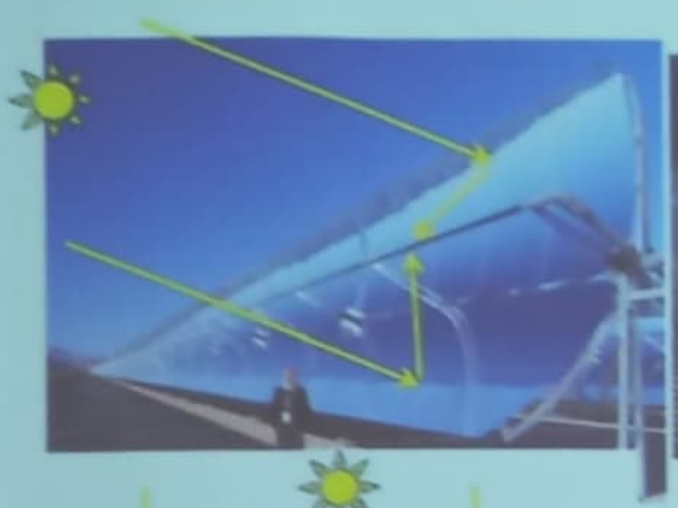
2 SEAGEN-F
Company: Minas Energy-Marine Current Turbines-Bluewater
Device: Floating dual-rotor turbine moored by cables
Size: 40 metres long, 16-metre rotors
Production: 2 MW
Deployment: 2016 or 2017

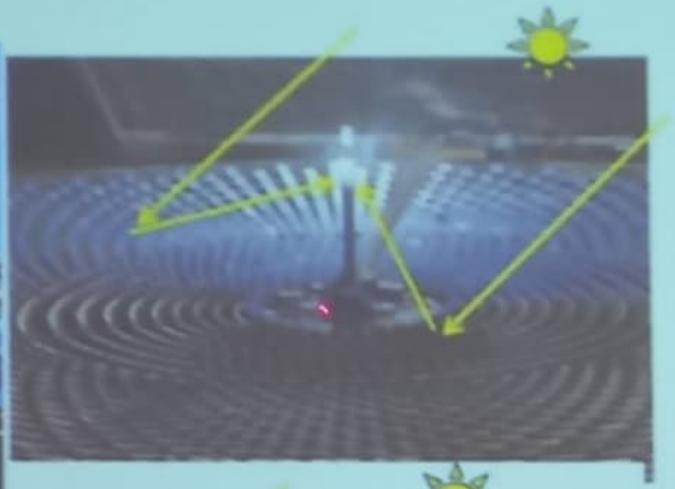
3 TRITON
Company: Black Rock Tidal Power
Device: Semi-submerged, 36-40 turbine array on a gravity base
Size: 46.6 metres long, 4-metre-diameter rotors
Production: 2.5 MW
Deployment: Fall 2016

4 AR1500
Company: Atlantis Established Marine Energy Shipbuilding
Device: Single-rotor turbine on a subsea foundation
Size: 18-metre-diameter rotor
Production: 1.5 MW
Deployment: 2017

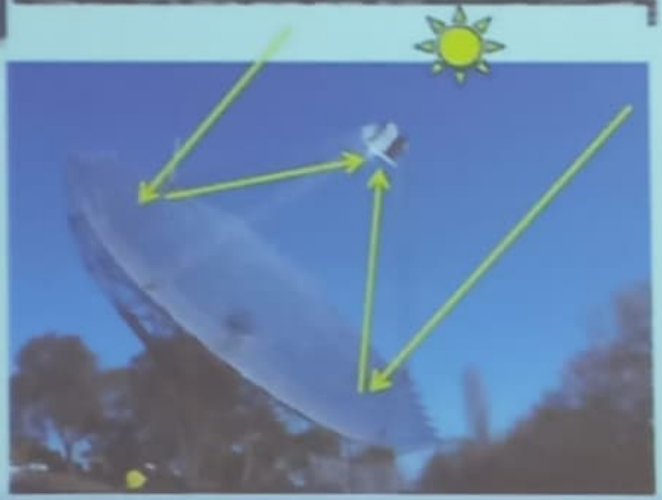
Types of Concentrating Solar Power

Four types: troughs, central tower receivers, linear Fresnel, dish







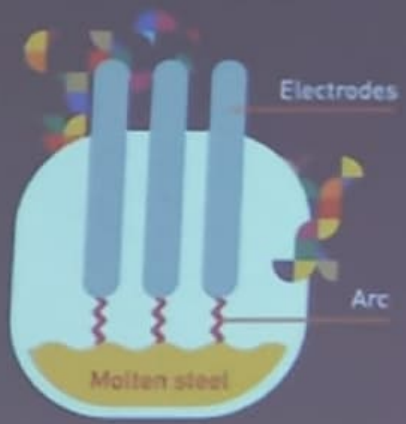


Trough & Fresnel (1 axis) concentration solar by 80 x

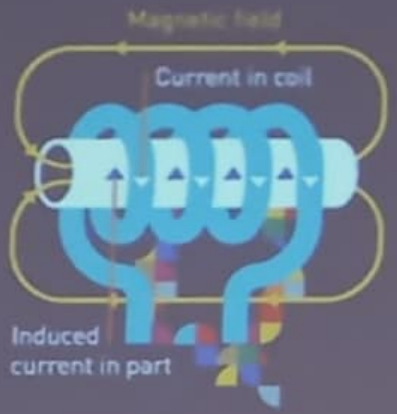
Tower and dish (2 axes) concentrate solar by 1000 x → higher temperature

HEATING PROCESSES

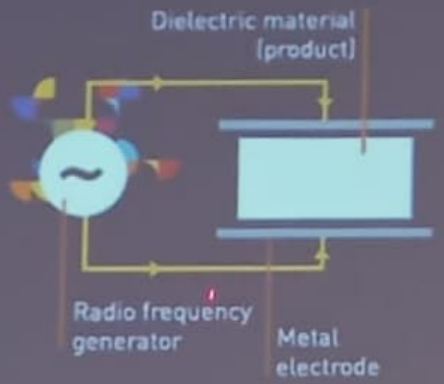
Electric arc heating



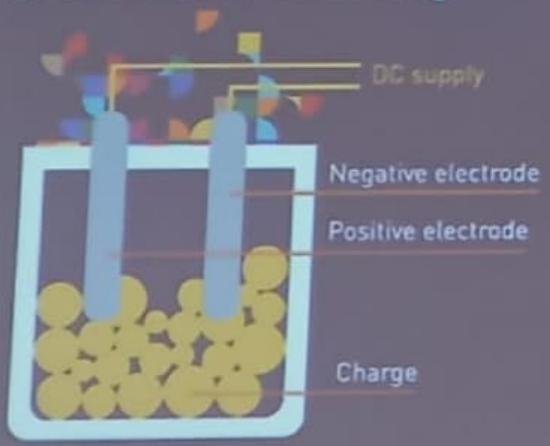
Induction heating



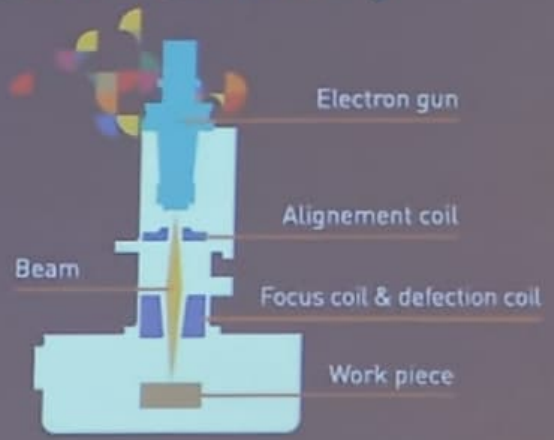
Dielectric heating



Direct resistance heating

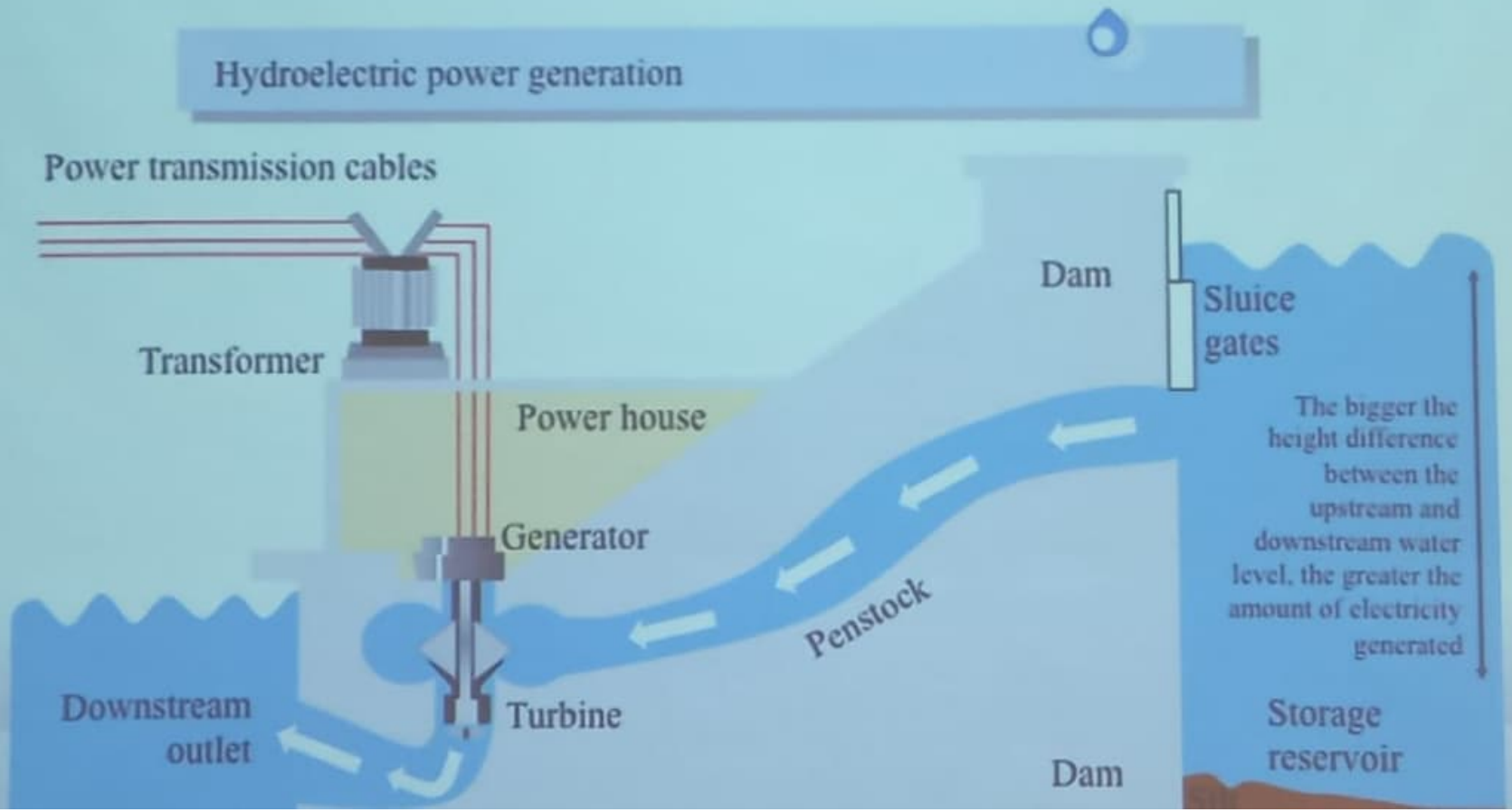


Electron beam heating



Hydroelectric Power as Storage

Water drains through turbine when electricity is needed; reservoir charges naturally through rainfall and runoff



Types of Storage for a 100% WWS System

ELECTRICITY	HEATING/COOLING	OTHER
CSP with storage	Water tank	Hydrogen
Pumped storage hydro	Ice	
Existing hydroelectric	Underground	
Batteries	Borehole	
Flywheels	Water Pit	
Compressed air	Aquifer	
Gravitational Storage	Building materials	